

# AI-NAV — Artificial Intelligence (CS-323) — Master Index

Weeks 1–15 · 17 Aug – 29 Nov 2026 · **Midterm 30** / **Final 40** / **Lab 20** · load 4.0 · difficulty 4/5

**Open this file first.** It is the map: where every topic lives, every line offset, every resource. **Four-file pack** (all in `terms/2026-fall/AI/`): | **File** | **Holds** | `---|---` | `AI-NAV.md` | This map — glance tables, master offsets, resources index | | `AI-BEFORE-MID.md` | **Weeks 1–8 (midterm)** — full per-week content, modules 1–6 | | `AI-AFTER-MID.md` | **Weeks 9–15 (final)** — full per-week content, modules 6–9 | | `AI-LABS.md` | All 13 labs + lab viva — lab→week map, per-lab blocks, viva prep |

## Where to look (fast answers)

| I want...                                     | Open                                                             |
|-----------------------------------------------|------------------------------------------------------------------|
| A topic with its defs/formulas/numericals     | that week's block in BEFORE-MID / AFTER-MID                      |
| Any week's exact line offsets                 | <b>Master offset table</b> below                                 |
| A full fear-killer pack (verbatim)            | that week's block — packs carry per-question offsets             |
| A lab, step-by-step                           | <code>AI-LABS.md</code> → per-lab block                          |
| Viva questions                                | <code>AI-LABS.md</code> viva section · <code>Viva-Book.md</code> |
| Books / YT playlists / MOOCs / practice banks | <b>Resources</b> section in this file                            |
| Exam-day order                                | <b>Exam stacks</b> section                                       |
| Which official module covers what             | <b>Syllabus M1–M9</b> section                                    |

## Course facts

- **Code:** CS-323 · **Credits:** 3+1 (theory + lab) · **Contact:** 3 lectures + 1 lab/week
- **Weightage:** Midterm 30% / Final 40% / Lab ~20% (canonical per `Marks-Allocation.md`; `01-Course-Overview.md` defers)
- **Difficulty:** 4/5 · **GPA Risk:** High
- **Rotation:** Course A (Wed + Fri) — 2 A-slots/wk · labs Saturday
- **Midterm:** Week 8 (05–11 Oct) · **Final:** Week 15 (23–29 Nov) · **Lab viva:** Week 15 (separate slot)
- **Exam character:** 40% of exam weight sits on three pillars — **A\*, GA, backpropagation**. Trace problems reward process over final number (≈70% partial credit).
- **Workbook note:** lab order runs **ANN-first** (W1–5), independently of the theory timeline (NN theory is W10–11). GA has **no lab** — theory pillar only.

## Master weekwise line-offset table

Offsets are line numbers into the named source file. `Fear-Killer-Packs.md` and `Week-by-Week-Narrative.md` are the two anchored sources; `weeks/AI-Wn.md` files live in `weeks/`. Only 8 packs exist (AI-W3/4/5/6/9/11/12/13) — W1/W2/W10 and W7/W14 (revision) have no pack.

| Week | Narrative | Pack heading                   | Pack questions                                                | weeks/AI-Wn | Module  |
|------|-----------|--------------------------------|---------------------------------------------------------------|-------------|---------|
| W1   | L18–33    | — (no pack)                    | —                                                             | AI-W1 L1–47 | M1      |
| W2   | L37–52    | — (no pack)                    | —                                                             | AI-W2 L1–46 | M2      |
| W3   | L56–74    | L11 (AI-W3: Search)            | res L12 · Q1 L14 · Q2 L15 · Q3 L16 · Q4 L17                   | AI-W3 L1–47 | M2      |
| W4   | L78–96    | L19 (AI-W4: CSP & Adversarial) | res L20 · Q1 L22 · Q2 L23 · Q3 L24 · Q4 L25                   | AI-W4 L1–48 | M2      |
| W5   | L100–116  | L27 (AI-W5: FOL & KR)          | res L28 · Q1 L30 · Q2 L31 · Q3 L32 · Q4 L34 · Q5 L35 · Q6 L36 | AI-W5 L1–47 | M3 + M5 |

|     |          |                                    |                                             |                          |       |
|-----|----------|------------------------------------|---------------------------------------------|--------------------------|-------|
| W6  | L120–141 | L38 (AI-W6: Genetic Algorithms)    | res L39 · Q1 L41 · Q2 L42 · Q3 L43 · Q4 L44 | AI-W6 L1–50              | M4    |
| W7  | L145–160 | — (revision)                       | —                                           | AI-W7 L1–56              | M1–M6 |
| W8  | L164–174 | — (no pack)                        | —                                           | AI-W8 L1–31              | —     |
| W9  | L177–197 | L46 (AI-W9: Rule-Based Systems)    | res L47 · Q1 L49 · Q2 L50 · Q3 L51          | AI-W9 L1–53              | M6    |
| W10 | L201–217 | — (no pack)                        | —                                           | AI-W10 L1–49             | M7    |
| W11 | L221–241 | L53 (AI-W11: Neural Networks)      | res L54 · Q1 L56 · Q2 L57 · Q3 L58 · Q4 L59 | AI-W11 L1–53             | M7    |
| W12 | L245–263 | L61 (AI-W12: Fuzzy Logic)          | res L62 · Q1 L64 · Q2 L65 · Q3 L66          | AI-W12 L1–51             | M8    |
| W13 | L268–283 | L68 (AI-W13: Game Theory & Ethics) | res L69 · Q1 L71 · Q2 L72 · Q3 L73 · Q4 L74 | AI-W13 L1–48             | M9    |
| W14 | L287–308 | — (taper)                          | —                                           | AI-W14 L1–56             | M1–M9 |
| W15 | L312–326 | — (no pack)                        | —                                           | (W15 note at AI-W14 L56) | —     |

## Syllabus M1–M9 → where it lives

| Module                                       | Breakdown?   | Covered in           |
|----------------------------------------------|--------------|----------------------|
| M1 Introduction & Computational Intelligence | ✗ pack-level | W1                   |
| M2 Searching Methodologies                   | ✗ pack-level | W2–W3                |
| M3 First-Order Logic                         | ✗ pack-level | W5                   |
| M4 Evolutionary Computing                    | ✗ pack-level | W6                   |
| M5 Knowledge Representation                  | ✗ pack-level | W5                   |
| M6 Imperfect Knowledge & Rule-Based Systems  | ✗ pack-level | W9                   |
| M7 Artificial Neural Networks                | ✗ pack-level | W10–W11 (labs W1–W5) |
| M8 Fuzzy Set Theory                          | ✗ pack-level | W12 (lab W14)        |
| M9 Game Theory & Prospects                   | ✗ pack-level | W13                  |

AI has no `03-Chapter-Breakdowns/` — all modules map at pack/narrative level (per `weeks/README.md`). Gaps are flagged, not fabricated. The official syllabus lists no per-module hours.

## Exam probability table (editorial — study prioritization only)

| Topic              | Midterm | Final | Numerical | Diagram | Theory | Definition |
|--------------------|---------|-------|-----------|---------|--------|------------|
| AI intro           | 70%     | 20%   | 0%        | 0%      | 60%    | 40%        |
| Uninformed search  | 90%     | 40%   | 25%       | 20%     | 35%    | 20%        |
| A* search          | 85%     | 55%   | 35%       | 25%     | 25%    | 15%        |
| CSP/Minimax        | 75%     | 40%   | 20%       | 25%     | 35%    | 20%        |
| FOL/Resolution     | 70%     | 50%   | 10%       | 15%     | 50%    | 25%        |
| Genetic algorithms | 40%     | 75%   | 35%       | 15%     | 30%    | 20%        |
| Expert systems     | 0%      | 60%   | 0%        | 15%     | 55%    | 30%        |

|                 |    |     |     |     |     |     |
|-----------------|----|-----|-----|-----|-----|-----|
| Neural networks | 0% | 80% | 25% | 20% | 35% | 20% |
| Backpropagation | 0% | 85% | 35% | 15% | 30% | 20% |
| Fuzzy logic     | 0% | 65% | 25% | 15% | 35% | 25% |

## Marks allocation

| Component          | Weight | Strategy                      |
|--------------------|--------|-------------------------------|
| A* search traces   | ~15%   | OPEN/CLOSED list practice     |
| Genetic Algorithms | ~13%   | One generation by hand drill  |
| Backpropagation    | ~12%   | 3-layer network, chain rule   |
| FOL/Resolution     | ~10%   | English-to-FOL conversions    |
| Fuzzy logic        | ~10%   | Mamdani inference trace       |
| Theory             | ~20%   | Flashcards                    |
| Lab                | ~20%   | Clean code + analysis + plots |

## Exam stacks

- **Midterm (W8, 05–11 Oct):** past paper 60 min → blank page 30 min → error log 20 min. Answer *A/search traces first, theory second*. A trace target < 15 min. Sleep 8 h. Ledger frozen.
- **Final (W15, 23–29 Nov):** A\* trace → GA → neural networks → FOL → fuzzy logic → theory. Write every step for trace problems — partial credit exists (process ≈ 70% of marks). Sleep 9 h (banked from W12).
- **Viva (W15, separate slot):** `AI-LABS.md` viva + `Viva-Book.md` — walkthrough first (3 min), then technical. Demonstrate your best implementation live; be ready to change a parameter and explain the expected effect.

## Resources

### YouTube playlists

| Playlist                                           | Channel         | Videos | Why                                                                                                  |
|----------------------------------------------------|-----------------|--------|------------------------------------------------------------------------------------------------------|
| <a href="#">Artificial Intelligence (Complete)</a> | Gate Smashers   | 60     | Search, FOL, genetic algorithms, knowledge representation, neural networks, fuzzy logic, game theory |
| <a href="#">Neural Networks (3Blue1Brown)</a>      | 3Blue1Brown     | ~4     | Best visual explanation of how neural networks work                                                  |
| <a href="#">Neural Networks: Zero to Hero</a>      | Andrej Karpathy | ~10    | From scratch in code. Build, train, and understand NNs deeply.                                       |

### Textbooks

- **Russell & Norvig** — *Artificial Intelligence: A Modern Approach*, 4th Ed (2020). ISBN **978-0-13-461099-3**. The AI bible; Ch.3 search, Ch.4 GA, Ch.5–6 CSP/adversarial, Ch.8–10 FOL/KR, Ch.18 game theory, Ch.21 NN, Ch.27 prospects.
- **AIMA Python code + AIMA Exercises** — [aima.cs.berkeley.edu](https://aima.cs.berkeley.edu) · [github.com/aimacode/aima-python](https://github.com/aimacode/aima-python)

### Free MOOCs

| Course                                                         | Source        | Notes                              |
|----------------------------------------------------------------|---------------|------------------------------------|
| <a href="#">NPTEL: Introduction to Artificial Intelligence</a> | IIT Kharagpur | Prof. Anupam Basu, Prof. S. Sarkar |
| <a href="#">MIT OCW: 6.034 Artificial Intelligence</a>         | MIT           | Patrick Winston's legendary course |
| <a href="#">edX: CS50's Introduction to AI with Python</a>     | Harvard       | Project-based intro                |

### Problem banks / practice

| Resource                              | Link                                                                                   |
|---------------------------------------|----------------------------------------------------------------------------------------|
| AIMA Exercises Platform               | https://aimacode.github.io/aima-exercises/                                             |
| AIMA Python Code                      | https://github.com/aimacode/aima-python                                                |
| UC Berkeley CS188 Pac-Man AI Projects | https://inst.eecs.berkeley.edu/~cs188/                                                 |
| MIT 6.034 Labs                        | https://ocw.mit.edu/courses/6-034-artificial-intelligence-fall-2010/pages/assignments/ |

Cheat sheets / revision notes

| Resource                               | Link                                                                                                                            |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Stanford CS 221 Super Cheatsheet (PDF) | https://github.com/afshinea/stanford-cs-221-artificial-intelligence/blob/master/en/super-cheatsheet-artificial-intelligence.pdf |
| McGill COMP424 Cheatsheet (PDF)        | https://www.cs.mcgill.ca/~yzhou53/stuff/comp424cheatsheet.pdf                                                                   |
| Cheatography: AI Cheat Sheet           | https://cheatsheetshero.com/user/all/161-artificial-intelligence-cheat-sheet                                                    |

Secret weapon

**AIMA Code Repository** — <https://github.com/aimacode/aima-python> — the official Python implementation of every algorithm in Russell & Norvig. Pair with [AIMA Exercises](#) for a complete self-study loop.

Where-everything-lives index (source → embedded location)

| Source file                         | ~Lines | Embedded in                                             |
|-------------------------------------|--------|---------------------------------------------------------|
| 01-Course-Overview.md               | 53     | NAV facts/weights + exam probability + marks allocation |
| 02-Official-Syllabus.md             | 86     | NAV (M1–M9 map, CLO, lab table + textbooks)             |
| 03-Lab-Breakdowns/01-ANN-and-ML     | 42     | AI-LABS (Labs 1–5, group block)                         |
| 03-Lab-Breakdowns/02-Searching      | 21     | AI-LABS (Labs 6–7, group block)                         |
| 03-Lab-Breakdowns/03-CSP            | 21     | AI-LABS (Labs 8–9, group block)                         |
| 03-Lab-Breakdowns/04-Expert-Systems | 28     | AI-LABS (Labs 10–12, group block)                       |
| 03-Lab-Breakdowns/05-Fuzzy-Logic    | 14     | AI-LABS (Lab 13, group block)                           |
| Definition-Book.md                  | 26     | per-week "Definitions" blocks (verbatim)                |
| Formula-Book.md                     | 38     | per-week "Formulas" blocks (verbatim)                   |
| Numerical-Book.md                   | 25     | per-week "Numericals" numbered types (#33–45)           |
| Diagram-Book.md                     | 21     | per-week "Diagrams" numbered refs (#1–15)               |
| Marks-Allocation.md                 | 14     | NAV (this file)                                         |
| Resources.md                        | 43     | NAV (this file) + per-week book/video lines             |
| Fear-Killer-Packs.md                | 77     | per-week packs (verbatim, with per-question offsets)    |
| Week-by-Week-Narrative.md           | 335    | per-week narrative (verbatim)                           |
| Lab-Schedule.md                     | 28     | NAV schedule + AI-LABS lab→week map                     |
| Lab-Resources.md                    | 140    | AI-LABS per-lab Resources lines                         |
| Viva-Book.md                        | 23     | AI-LABS viva section (verbatim)                         |
| weeks/AI-W1...W14 + README          | 682    | per-week anchors, P0 floor, drill targets               |

Top-10-Tricky-Concepts.md + Top-100-Questions.md are **deferred** for AI (decision 2026-08-04) — equivalent coverage lives in Definition-Book.md, Fear-Killer-Packs.md, and the topical books.

## Schedule at a glance

| W  | Dates         | Variant | File   | Topics                                 | Module  | Lab    |
|----|---------------|---------|--------|----------------------------------------|---------|--------|
| 1  | 17–23 Aug     | P2      | BEFORE | Intro to AI & problem solving          | 01      | Lab 1  |
| 2  | 24–30 Aug     | P2      | BEFORE | Uninformed search                      | 02      | Lab 2  |
| 3  | 31 Aug–06 Sep | P2      | BEFORE | Informed search & A*                   | 02      | Lab 3  |
| 4  | 07–13 Sep     | P0      | BEFORE | CSPs & adversarial search              | 02      | Lab 4  |
| 5  | 14–20 Sep     | P0      | BEFORE | FOL & knowledge representation         | 03 + 05 | Lab 5  |
| 6  | 21–27 Sep     | P2      | BEFORE | Genetic algorithms                     | 04      | Lab 6  |
| 7  | 28 Sep–04 Oct | P1      | BEFORE | Midterm revision                       | 01–06   | Lab 7  |
| 8  | 05–11 Oct     | MIDTERM | BEFORE | Exam week — no new material            | —       | —      |
| 9  | 12–18 Oct     | P1      | AFTER  | Rule-based & expert systems (recovery) | 06      | Lab 8  |
| 10 | 19–25 Oct     | P0      | AFTER  | Neural networks I: perceptron          | 07      | Lab 9  |
| 11 | 26 Oct–01 Nov | P2      | AFTER  | Neural networks II: backpropagation    | 07      | Lab 10 |
| 12 | 02–08 Nov     | P0      | AFTER  | Fuzzy logic                            | 08      | Lab 11 |
| 13 | 09–15 Nov     | P1      | AFTER  | Game theory & ethics                   | 09      | Lab 12 |
| 14 | 16–22 Nov     | P0      | AFTER  | Final taper — past paper FIRST         | 01–09   | Lab 13 |
| 15 | 23–29 Nov     | FINAL   | AFTER  | Exam + lab viva — execute              | —       | —      |

Tier key: **P0** = no exam pressure (front-load new material) · **P1** = light revision · **P2** = drill-heavy week. Variants are editorial tier assignments mirroring CA/PS — AI manifests carry no variant field. Lab column follows Lab-Schedule.md (workbook order; W8/W15 no lab).